Assignment 1: Random Projections

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- *Course: Machine Learning 2*
- *Prof: Jan Nagler*
- *Due: 17 April 2026*

Methodology

In order to complete the assignment we structure the task in seven steps. The first five (1-5) are done for each dataset. In step 6 we compare the results of both datasets to eachother and finally in step 7 we summarize our findings and conclude with our learnings and take-aways.

In this section we provide a brief description of each step:

Step 1 - Loading the Datasets

Dataset 1: COIL-20 (Columbia Object Image Library)

- `1,440` grayscale images of `20` objects, each photographed from `72` rotation angles (5° intervals)
- Each image is `128x128` pixels, giving **16,384 features** per sample when flattened
- Labels correspond to object identity (20 classes)
- Features normalized to [0, 1] by dividing pixel values by 255

Why this dataset:

- Images live in Euclidean space where pixel-to-pixel distance is a valid similarity measure.
- The data is very high-dimensional yet structurally low-dimensional (smooth object rotations form a manifold), making it an ideal candidate for random projections.

Dataset 2: Breast Cancer Wisconsin (Diagnostic)

- `569` samples labeled as malignant or benign (binary classification)
- Each sample has **30 numerical features** derived from digitized cell nucleus images (radius, texture, smoothness, concavity, etc.)
- Features standardized using `StandardScaler` (zero mean, unit variance)

Why this dataset:

- It serves as a deliberate contrast to COIL-20.
- The feature space is already low-dimensional and tabular rather than image-based, letting us test whether random projections still preserve useful structure when no clear geometric manifold exists.

Step 2 - Johnson-Lindenstrauss Bound

The JL lemma gives a theoretical lower bound on the number of dimensions k needed to approximately preserve all pairwise distances:

$$k \geq \frac{4\ln(n)}{\frac{\varepsilon^2}{2} - \frac{\varepsilon^3}{3}}$$

- n = number of samples, ε = allowable distortion (we use $\varepsilon = 0.1$)
- Computed via `johnson_lindenstrauss_min_dim` from scikit-learn
- Acts as a reference point; the empirically required k is often far smaller

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Step 3 - Baseline Accuracy

- A `LinearSVC` is trained on the **original, unprojected** data
- Accuracy on the test set defines the **baseline** - the performance ceiling we aim to recover after projection
- Fully independent of random projections; purely benchmarks the classifier

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Step 4 - Sparse Random Projection (Single Run)

- A sparse random matrix is constructed where most entries are 0 and non-zero entries are drawn from $\{-1, +1\}$
- Data is projected to a target dimension k, then a fresh `LinearSVC` is trained and evaluated

- Repeated across a range of k values (2 to 500 for COIL-20; 2 to 30 for Breast Cancer)

Note: Using a single random draw can produce irregular dips caused by the particular projection matrix chosen.

Step 5 - Ensembled Random Projection

- For each k, **25 independent projections** are run with different random seeds
- A separate `LinearSVC` is trained and evaluated per projection
- The 25 scores are averaged into a single stable value per k

Why this matters:

- Averaging over multiple projections removes noise from any single random matrix, producing the smooth and reliable accuracy curve that is the main deliverable of this analysis.

Step 6 - Comparison of Both Datasets

- Ensembled accuracy curves for COIL-20 and Breast Cancer plotted side by side
- x-axis normalized to fraction of max projected dimensions ($k/k_{\max}$) so both datasets can be overlaid
- A second panel shows the **accuracy gap** (baseline minus projected accuracy) across relative dimensionality
- Summary table reports baseline accuracy, best projected accuracy, and $k_{\max}$ for each dataset

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Step 7 - Conclusion

Summarizes the trade-offs between dimensionality reduction and classification performance, and discusses when random projection is most effective based on the observed results.

Importing necessary libraries

```
from sklearn.random_projection import SparseRandomProjection
from sklearn.random_projection import johnson_lindenstrauss_min_dim
from sklearn.svm import LinearSVC
from sklearn.model_selection import train_test_split
from sklearn import metrics
```

```
import numpy as np
import matplotlib.pyplot as plt
import warnings
import io, urllib, os
import zipfile
from PIL import Image
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
warnings.filterwarnings('ignore')
%matplotlib inline
np.random.seed(42)
```

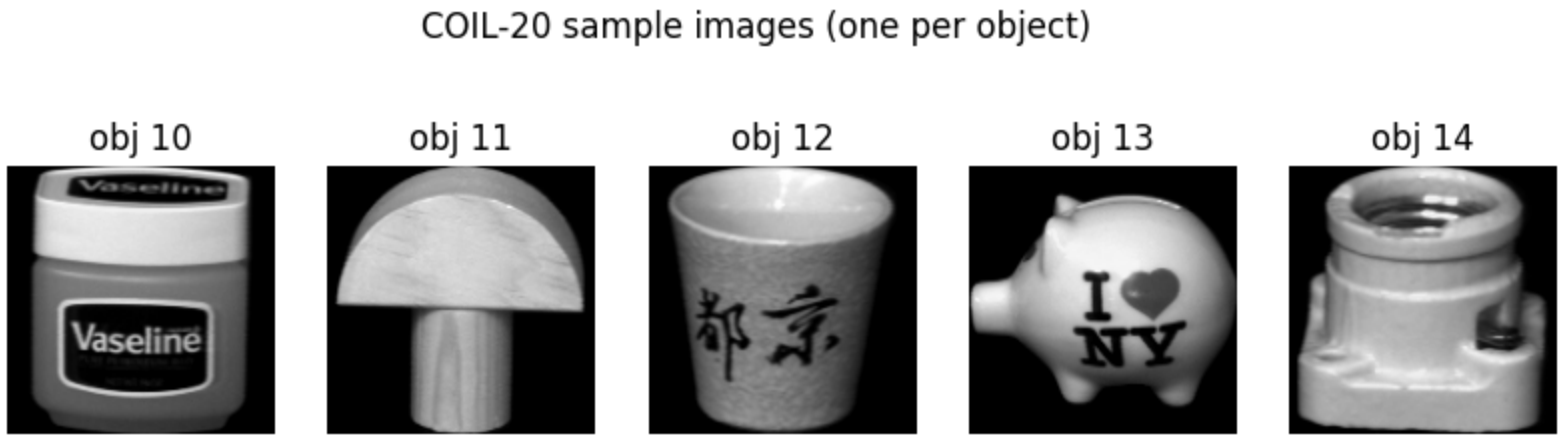
Dataset 1: Columbia Object Image Library dataset (COIL-20)

1. Loading the dataset

```
# Downloading the COIL-20 dataset
response = urllib.request.urlopen(
    'http://www.cs.columbia.edu/CAVE/databases/SLAM_coil-20_coil-100/coil-20/coil-20-proc.zip')
zip_bytes = io.BytesIO(response.read())
X_coil, y_coil = [], []
with zipfile.ZipFile(zip_bytes) as zf:
    for fname in sorted(zf.namelist()):
        if fname.endswith('.png'):
            label = int(fname.split('obj')[1].split('_')[0]) - 1
            img_bytes = io.BytesIO(zf.read(fname))
            img = Image.open(img_bytes).convert('L')
            X_coil.append(np.array(img).flatten().astype('float32') / 255.0)
            y_coil.append(label)
X_coil = np.array(X_coil)
y_coil = np.array(y_coil)
print('COIL-20 shape:', X_coil.shape)
print('Number of classes:', len(np.unique(y_coil)))
```

```
COIL-20 shape: (1440, 16384)
Number of classes: 20
```

```
# Displaying some sample images from the dataset
fig, axes = plt.subplots(1, 5, figsize=(10, 3))
for i, ax in enumerate(axes):
    ax.imshow(X_coil[i * 72].reshape(128, 128), cmap='gray')
    ax.set_title('obj %d' % (y_coil[i * 72] + 1))
    ax.axis('off')
plt.suptitle('COIL-20 sample images (one per object)')
plt.show()
```




```
# Splitting the dataset into training and testing sets (70% train, 30% test)
X_coil_train, X_coil_test, y_coil_train, y_coil_test = train_test_split(X_coil, y_coil,
                                                                        test_size=0.3, random_state=42)
print('Train size:', X_coil_train.shape)
print('Test size: ', X_coil_test.shape)
print('Feature dimension (original):', X_coil_train.shape[1])
```

```
Train size: (1008, 16384)
Test size:  (432, 16384)
Feature dimension (original): 16384
```

2. Johnson and Lindenstrauss Bound

```
# Johnson-Lidenstrauss bound for COIL-20 (eps=0.1)
# For n=1440 samples, dimension d=128x128=16384 (pixels)
print('Professors Johnson and Lindenstrauss say: k >=',johnson_lindenstrauss_min_dim(len(X_coil), eps=0.1))
```

```
Professors Johnson and Lindenstrauss say: k >= 6233
```

3. Getting baseline accuracy

```
model_coil=LinearSVC(max_iter=5000)
model_coil.fit(X_coil_train, y_coil_train)
baseline_coil=metrics.accuracy_score(y_coil_test, model_coil.predict(X_coil_test))
print('Baseline accuracy (no projection): %.4f' % baseline_coil)
```

```
Baseline accuracy (no projection): 0.9907
```

4. Sparse Random Projection

```
# Applying sparse random projection for varying dimensions
accuracies_coil = []
dims_coil = np.int32(np.linspace(2, 500, 20))
for dim in dims_coil:
    sp = SparseRandomProjection(n_components=dim)
    X_proj = sp.fit_transform(X_coil_train)
    model = LinearSVC(max_iter=5000)
    model.fit(X_proj, y_coil_train)
    test_proj = sp.transform(X_coil_test)
    accuracies_coil.append(metrics.accuracy_score(y_coil_test, model.predict(test_proj)))
```

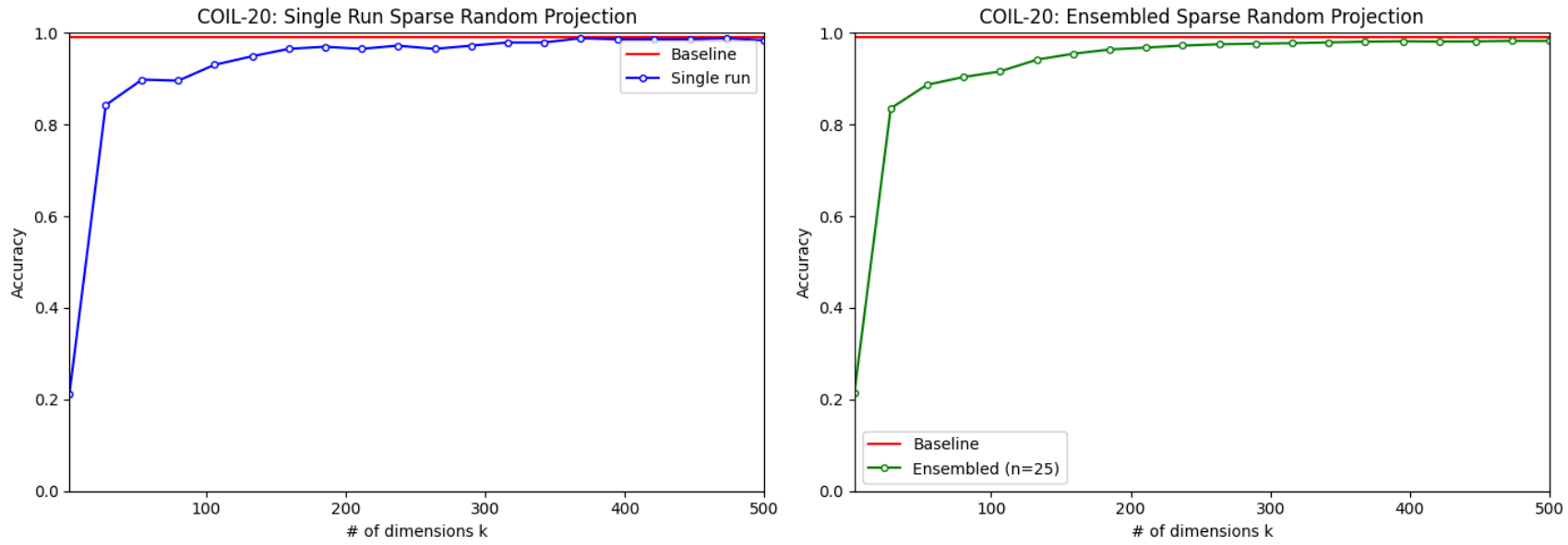
5. Ensembled Random Projection

```
# Applying ensembled sparse random projection over multiple realizations
n_realizations = 25
accuracies_coil_ens = np.zeros((n_realizations, len(dims_coil)))
for j, dim in enumerate(dims_coil):
    for i in range(n_realizations):
```

```
        sp = SparseRandomProjection(n_components=dim)
        X_proj = sp.fit_transform(X_coil_train)
        model = LinearSVC(max_iter=5000)
        model.fit(X_proj, y_coil_train)
        test_proj = sp.transform(X_coil_test)
        accuracies_coil_ens[i, j] = metrics.accuracy_score(y_coil_test, model.predict(test_proj))
# Averaging across realizations
avg_accuracies_coil = accuracies_coil_ens.mean(axis=0)
```

Plotting Sparse Random Projection and Ensembled random projection

```
# Plotting single run sparse random projection
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].plot(dims_coil, [baseline_coil] * len(dims_coil), color='r', label='Baseline')
axes[0].plot(dims_coil, accuracies_coil, color='b', label='Single run',marker='o',
             markersize=4, markerfacecolor='white', markeredgewidth=1)
axes[0].set_xlabel('# of dimensions k')
axes[0].set_ylabel('Accuracy')
axes[0].set_title('COIL-20: Single Run Sparse Random Projection')
axes[0].set_xlim([2, 500])
axes[0].set_ylim([0, 1])
axes[0].legend()
# Plotting ensembled sparse random projection
axes[1].plot(dims_coil, [baseline_coil] * len(dims_coil), color='r', label='Baseline')
axes[1].plot(dims_coil, avg_accuracies_coil, color='g', label='Ensembled (n=25)',
             marker='o', markersize=4, markerfacecolor='white', markeredgewidth=1)
axes[1].set_xlabel('# of dimensions k')
axes[1].set_ylabel('Accuracy')
axes[1].set_title('COIL-20: Ensembled Sparse Random Projection')
axes[1].set_xlim([2, 500])
axes[1].set_ylim([0, 1])
axes[1].legend()
plt.tight_layout()
plt.show()
```



Conclusion for Dataset 1

- The baseline accuracy without any projection is `0.9907` (99.07%), computed using a LinearSVC on the full `16,384`-dimensional space.
- The *single run curve* rises steeply from low k and increases smoothly toward the baseline, reflecting typical behavior for a single random projection draw.
- The *ensembled curve* (25 realizations averaged) produces a stable, reliable accuracy estimate, confirming that averaging over multiple random projections removes variance from any single projection matrix.
- Near-baseline accuracy (`~98%`) is recovered at roughly `k=300–400` dimensions, far below the JL bound of 6233, suggesting the bound is highly conservative for this dataset.

- This result aligns with the expected behavior for image data embedded in Euclidean space: since pixel-wise Euclidean distance is a meaningful similarity measure for COIL-20, random projections preserve the structure needed for classification even at a small fraction of the original dimensionality.

Dataset 2: Breast Cancer Wisconsin (Diagnostic)

1. Loading the dataset

```
from sklearn.datasets import load_breast_cancer
data_bc = load_breast_cancer(as_frame=True)
df_bc= data_bc.frame
X_bc = df_bc.drop(columns='target').values
y_bc = df_bc['target'].values
print('Breast Cancer shape:', X_bc.shape)
print('Number of classes:', len(np.unique(y_bc)))
```

```
Breast Cancer shape: (569, 30)
Number of classes: 2
```

```
print(df_bc.columns)
df_bc.sample(4)
```

```
Index(['mean radius', 'mean texture', 'mean perimeter', 'mean area',
       'mean smoothness', 'mean compactness', 'mean concavity',
       'mean concave points', 'mean symmetry', 'mean fractal dimension',
       'radius error', 'texture error', 'perimeter error', 'area error',
       'smoothness error', 'compactness error', 'concavity error',
       'concave points error', 'symmetry error', 'fractal dimension error',
       'worst radius', 'worst texture', 'worst perimeter', 'worst area',
       'worst smoothness', 'worst compactness', 'worst concavity',
       'worst concave points', 'worst symmetry', 'worst fractal dimension',
       'target'],
      dtype='str')
```

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	mean concave points	mean symmetry	mean fractal dimension	...	t
468	17.600	23.33	119.00	980.5	0.09289	0.20040	0.21360	0.10020	0.1696	0.07369	...	
357	13.870	16.21	88.52	593.7	0.08743	0.05492	0.01502	0.02088	0.1424	0.05883	...	
521	24.630	21.60	165.50	1841.0	0.10300	0.21060	0.23100	0.14710	0.1991	0.06739	...	
358	8.878	15.49	56.74	241.0	0.08293	0.07698	0.04721	0.02381	0.1930	0.06621	...	

4 rows × 31 columns

```
# Train-test split (70% train, 30% test)
X_bc_train, X_bc_test, y_bc_train, y_bc_test = train_test_split(X_bc, y_bc,
                                                                test_size=0.3, random_state=42, stratify=y_bc)
print('Train size:', X_bc_train.shape)
print('Test size: ', X_bc_test.shape)
print('Feature dimension (original):', X_bc_train.shape[1])
```

```
Train size: (398, 30)
Test size:  (171, 30)
Feature dimension (original): 30
```

```
# Scaling the features
scaler = StandardScaler()
```

```
X_bc_train_scaled = scaler.fit_transform(X_bc_train)
X_bc_test_scaled = scaler.transform(X_bc_test)
```

2. Johnson and Lindenstrauss Bound

```
# Johnson-Lindenstrauss bound for Breast Cancer (eps=0.1)
print('Professors Johnson and Lindenstrauss say: k >=',
      johnson_lindenstrauss_min_dim(len(X_bc), eps=0.1))
```

```
Professors Johnson and Lindenstrauss say: k >= 5437
```

- Although the dataset has only `30` features, the required dimension is larger than `5437`.
- The required embedding dimension depends only on the **number of samples** `(n=569)`. The large number occurs because the bound is a **conservative worst-case guarantee for preserving all pairwise distances**, and can therefore exceed the original dimensionality. In practice, much lower dimensions are often sufficient.

3. Getting baseline accuracy

```
model_bc = LinearSVC(max_iter=5000)
model_bc.fit(X_bc_train_scaled, y_bc_train)
baseline_bc = metrics.accuracy_score(y_bc_test, model_bc.predict(X_bc_test_scaled))
print('Baseline accuracy (no projection): %.4f' % baseline_bc)
```

```
Baseline accuracy (no projection): 0.9649
```

4. Sparse Random Projection

```
accuracies_bc = []
dims_bc = np.int32(np.linspace(2, 30, 15))
for dim in dims_bc:
    sp = SparseRandomProjection(n_components=dim)
    X_proj = sp.fit_transform(X_bc_train_scaled)
```

```
    model = LinearSVC(max_iter=5000)
    model.fit(X_proj, y_bc_train)
    test_proj = sp.transform(X_bc_test_scaled)
    accuracies_bc.append(metrics.accuracy_score(y_bc_test, model.predict(test_proj)))
```

5. Ensembled Random Projection

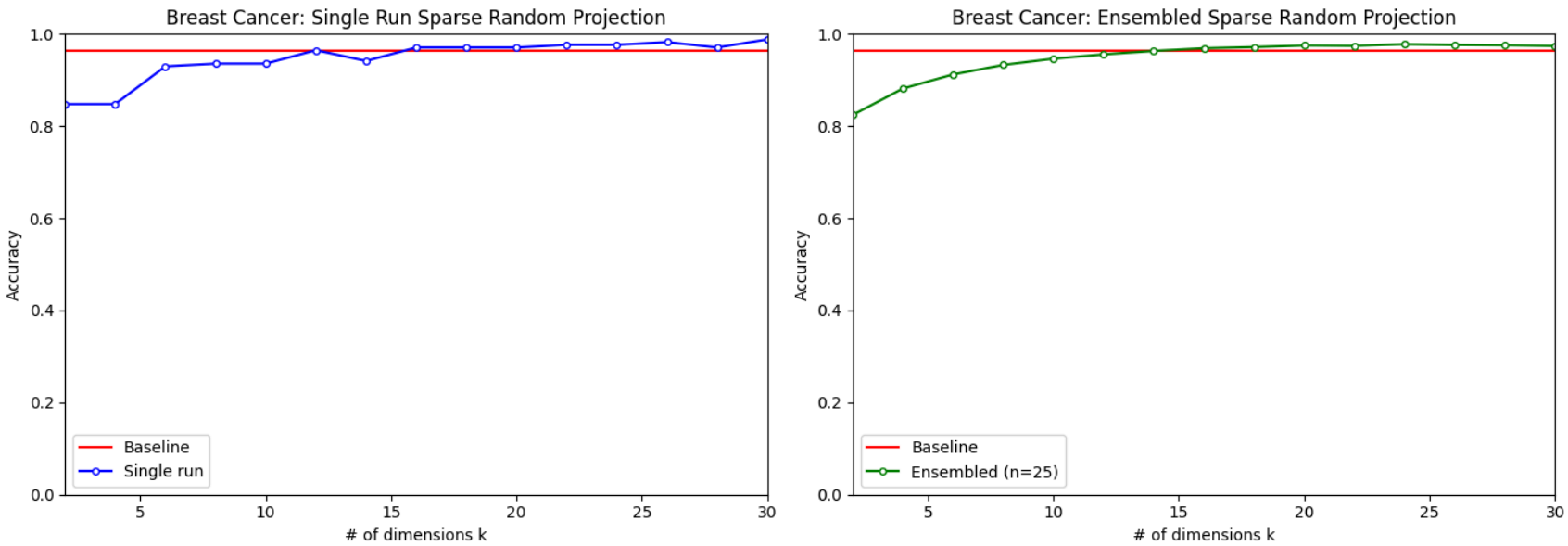
```
n_realizations = 25
accuracies_bc_ens = np.zeros((n_realizations, len(dims_bc)))
for j, dim in enumerate(dims_bc):
    for i in range(n_realizations):
        sp = SparseRandomProjection(n_components=dim)
        X_proj = sp.fit_transform(X_bc_train_scaled)
        model = LinearSVC(max_iter=5000)
        model.fit(X_proj, y_bc_train)
        test_proj = sp.transform(X_bc_test_scaled)
        accuracies_bc_ens[i, j] = metrics.accuracy_score(
            y_bc_test, model.predict(test_proj)
        )
avg_accuracies_bc = accuracies_bc_ens.mean(axis=0)
```

Plotting Sparse Random Projection and Ensembled random projection

```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
# Plotting single run sparse random projection
axes[0].plot(dims_bc, [baseline_bc] * len(dims_bc), color='r', label='Baseline')
axes[0].plot(dims_bc, accuracies_bc, color='b', label='Single run',
             marker='o', markersize=4, markerfacecolor='white', markeredgewidth=1)
axes[0].set_xlabel('# of dimensions k')
axes[0].set_ylabel('Accuracy')
axes[0].set_title('Breast Cancer: Single Run Sparse Random Projection')
axes[0].set_xlim([2, 30])
axes[0].set_ylim([0, 1])
axes[0].legend()
# Plotting ensembled sparse random projection
```

```
axes[1].plot(dims_bc, [baseline_bc] * len(dims_bc), color='r', label='Baseline')
axes[1].plot(dims_bc, avg_accuracies_bc, color='g', label='Ensembled (n=25)',
             marker='o', markersize=4, markerfacecolor='white', markeredgewidth=1)
axes[1].set_xlabel('# of dimensions k')
axes[1].set_ylabel('Accuracy')
axes[1].set_title('Breast Cancer: Ensembled Sparse Random Projection')
axes[1].set_xlim([2, 30])
axes[1].set_ylim([0, 1])
axes[1].legend()

plt.tight_layout()
plt.show()
```



Conclusion for Dataset 2

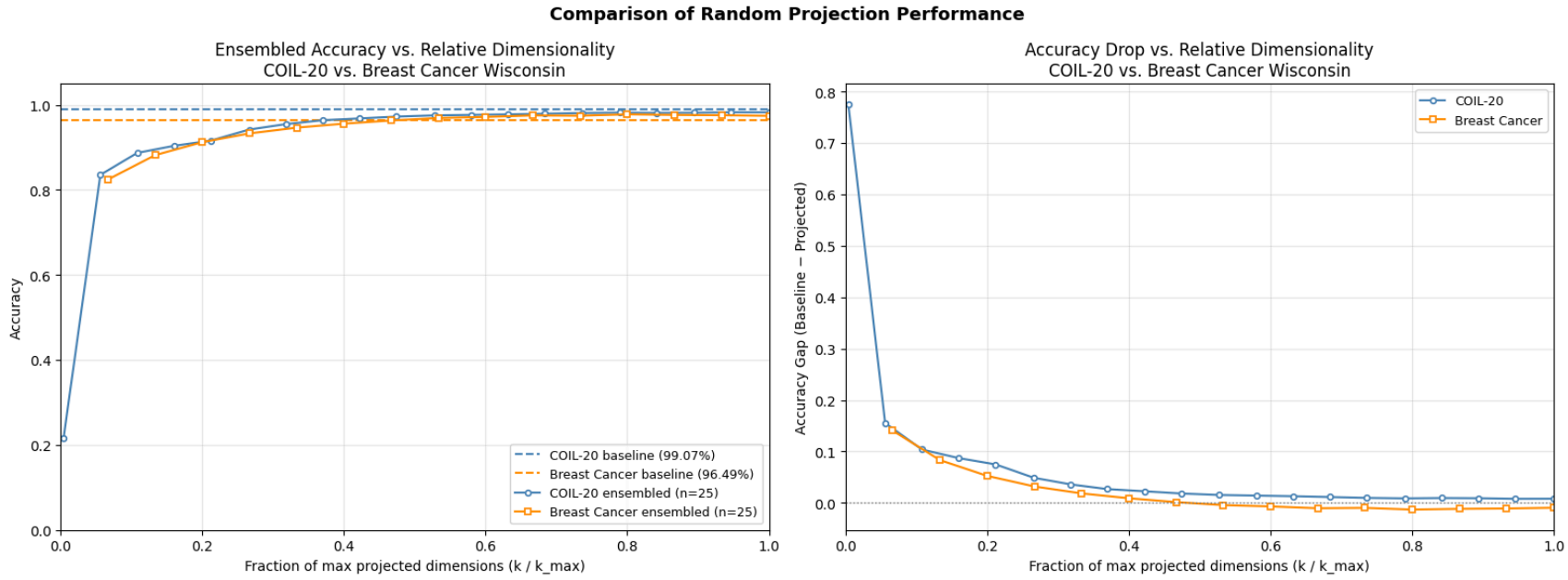
- The baseline accuracy without any projection is `0.9649` (96.49%), computed using a LinearSVC on the original 30-dimensional feature space.
- The *single run curve* shows noticeable fluctuations across k values, including a significant drop at very low k `(k=2–3)` and occasional dips around `k=14–15`. These variations are expected, as a single random projection matrix may distort pairwise distances differently at each dimensionality.
- The *ensembled curve* (25 realizations averaged) is **visibly smoother**, again confirming that averaging over multiple random projections removes variance caused by any single projection matrix.
- High accuracy is preserved even at relatively low dimensions `(e.g., k≈15–20)`, indicating that the dataset has a **low intrinsic dimensionality** and does not require all 30 original features to achieve strong performance.

6. Comparing COIL-20 and Breast Cancer Wisconsin Datasets

```
# Normalising x-axis to [0, 1] for comparison
x_coil_norm = dims_coil / dims_coil.max()
x_bc_norm   = dims_bc   / dims_bc.max()
fig, axes = plt.subplots(1, 2, figsize=(16, 6))
# Left panel: Ensembled accuracy curves (both datasets)
axes[0].axhline(baseline_coil, color='steelblue',  linestyle='--', linewidth=1.5,
                label=f'COIL-20 baseline ({baseline_coil:.2%})')
axes[0].axhline(baseline_bc,   color='darkorange', linestyle='--', linewidth=1.5,
                label=f'Breast Cancer baseline ({baseline_bc:.2%})')
axes[0].plot(x_coil_norm, avg_accuracies_coil,
             color='steelblue',  marker='o', markersize=4,
             markerfacecolor='white', markeredgewidth=1.2,
             label='COIL-20 ensembled (n=25)')
axes[0].plot(x_bc_norm, avg_accuracies_bc,
             color='darkorange', marker='s', markersize=4,
             markerfacecolor='white', markeredgewidth=1.2,
             label='Breast Cancer ensembled (n=25)')
axes[0].set_xlabel('Fraction of max projected dimensions (k / k_max)')
axes[0].set_ylabel('Accuracy')
axes[0].set_title('Ensembled Accuracy vs. Relative Dimensionality\nCOIL-20 vs. Breast Cancer Wisconsin')
axes[0].set_xlim([0, 1])
axes[0].set_ylim([0, 1.05])
```

```
axes[0].legend(loc='lower right', fontsize=9)
axes[0].grid(alpha=0.3)

# Right panel: Accuracy gap to baseline
gap_coil = baseline_coil - avg_accuracies_coil
gap_bc   = baseline_bc   - avg_accuracies_bc
axes[1].plot(x_coil_norm, gap_coil,
             color='steelblue',  marker='o', markersize=4,
             markerfacecolor='white', markeredgewidth=1.2,
             label='COIL-20')
axes[1].plot(x_bc_norm, gap_bc,
             color='darkorange', marker='s', markersize=4,
             markerfacecolor='white', markeredgewidth=1.2,
             label='Breast Cancer')
axes[1].axhline(0, color='grey', linestyle=':', linewidth=1)
axes[1].set_xlabel('Fraction of max projected dimensions (k / k_max)')
axes[1].set_ylabel('Accuracy Gap (Baseline – Projected)')
axes[1].set_title('Accuracy Drop vs. Relative Dimensionality\nCOIL-20 vs. Breast Cancer Wisconsin')
axes[1].set_xlim([0, 1])
axes[1].legend(loc='upper right', fontsize=9)
axes[1].grid(alpha=0.3)
plt.suptitle('Comparison of Random Projection Performance', fontsize=13, fontweight='bold')
plt.tight_layout()
plt.show()
# Summary table
print(f"\n{'-'*70}")
print(f"{'Dataset':<30} {'Baseline':>10} {'Best projected':>15} {'k_max':>6}")
print(f"{'-'*70}")
print(f"{'COIL-20':<30} {baseline_coil:>10.2%} {avg_accuracies_coil.max():>15.2%} {dims_coil.max():>6d}")
print(f"{'Breast Cancer Wisconsin':<30} {baseline_bc:>10.2%} {avg_accuracies_bc.max():>15.2%} {dims_bc.max():>
print(f"{'-'*70}")
```




```
----------------------------------------------------------------------
Dataset                          Baseline  Best projected  k_max
----------------------------------------------------------------------
COIL-20                            99.07%          98.27%    500
Breast Cancer Wisconsin            96.49%          97.78%     30
----------------------------------------------------------------------
```

7. Conclusion

Comparison

	COIL-20	Breast Cancer
Original dimensionality	16,384	30

	COIL-20	Breast Cancer
Baseline accuracy	99.07%	96.49%
Best projected accuracy	~98%	~97–98%
Dimensions needed to recover baseline	~300–400	~15–20
Smoothness of ensembled curve	High	High

Final Comparative Findings

- The comparison highlights how random projection behaves differently depending on the dimensionality and geometric structure of the dataset.
- For **COIL-20**, a high-dimensional image dataset, random projection achieves over `97%` dimensionality reduction with minimal loss in accuracy (~0.6 below baseline), confirming that the data lies on a *low-dimensional manifold where Euclidean distances are meaningful and well-preserved under projection.*
- For the **Breast Cancer dataset**, the ensembled accuracy curve rises steadily from low k and approaches baseline at around `k=15–20` dimensions out of `30`. Notably, at higher k the projected accuracy slightly exceeds the baseline, suggesting a mild regularization effect from projection. This shows that even in a low-dimensional tabular setting, random projection effectively preserves the discriminative structure of the data.
- These results demonstrate that *random projection achieves the most dramatic compression gains for high-dimensional data with underlying geometric structure*. For *low-dimensional tabular data, compression gains are more modest but classification performance is still well-preserved.*

Take-Home Message

- Random projection is most effective when the data lies in a **high-dimensional space with low intrinsic dimensionality and when Euclidean distances meaningfully capture similarity**.
- Its practical benefit depends less on the nominal dimensionality and more on the underlying geometric structure of the data.

- While random projections can preserve classification performance across different settings, substantial dimensionality reduction is achievable only when significant redundancy exists in the feature space.
- Finally, **averaging over multiple random projections is essential for reducing variance and obtaining stable, reliable performance estimates.**